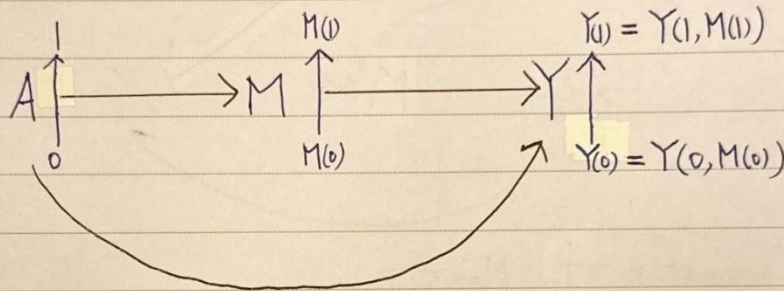


We introduce potential outcomes notation of Y and M

$M(a)$: potential outcome of mediator M to assigning treatment $A=a$

$Y(a, m)$: potential outcome for the outcome Y to assigning treatment $A=a$ and mediator $M=m$

$Y(a, M(a^*))$: that is, assign $A=a$ and then assign the mediator M to its potential outcome had we assigned treatment $A=a^*$



The total effect (TE)

$Y(1)$ vs $Y(0)$

$$TE = E[Y(1) - Y(0)] \quad IE: M(0) \rightarrow M(1)$$

$$= E[Y(1, M(1)) - Y(0, M(0))]$$

DE: $0 \rightarrow 1$

$$= \{E[Y(1, M(1))] - E[Y(1, M(0))]\} \quad NIE$$

$$+ \{E[Y(1, M(0))] - E[Y(0, M(0))]\} \quad NDE$$

差在
Interaction

Another decomposition

$$TE = \{E[Y(1, M(0))] - E[Y(0, M(1))]\} \quad TDE$$

$$+ \{E[Y(0, M(1))] - E[Y(0, M(0))]\} \quad PIE$$

$$TE = NDE + NIE = TDE + PIE$$

★ Mediation parameter

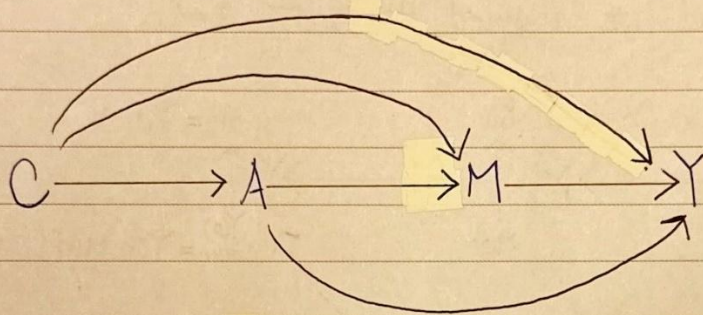
$$\phi(a, a^*) = E[Y(a, M(a^*))]$$

then

$$TE = \phi(1,1) - \phi(0,0)$$

$$NIE = \phi(1,1) - \phi(1,0)$$

$$NDE = \phi(1,0) - \phi(0,0)$$



★ Identification assumption in Mediation

1. Consistency

When $A=a$, the counterfactual outcome $Y(a)$ and $M(a)$ are, respectively, equal to the observed outcome Y and M

$$\begin{cases} Y(a) = Y \\ M(a) = M \end{cases} \text{ if } A=a$$

When $A=a$ and $M=m$, the counterfactual outcome $Y(a,m)$ is equal to observed Y

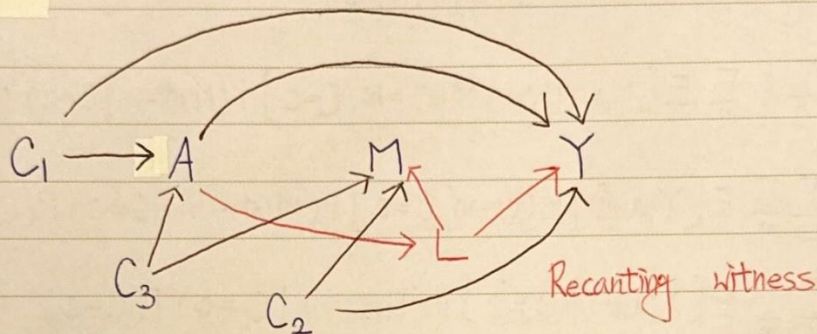
$$Y(a,m) = Y \text{ if } \begin{cases} A=a \\ M=m \end{cases}$$

2. Composition

The value of Y that would occur if A were set to a is equal to the value of Y that would occur if A were set to a and M were set to what it would have been if A were set to a .

$$Y(a) = Y(a, M(a))$$

3. Exchangeability



i) $Y(a, m) \perp\!\!\!\perp A \mid C$ No unmeasured A-Y confounding

ii) $Y(a, m) \perp\!\!\!\perp M \mid A, C$ No unmeasured M-Y confounding

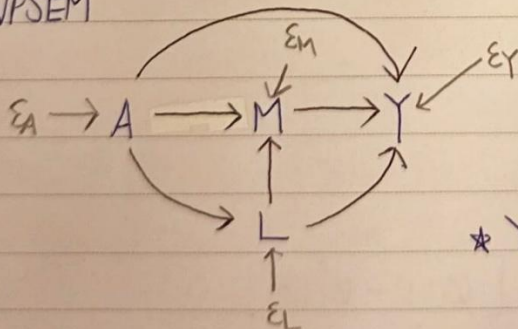
iii) $M(a) \perp\!\!\!\perp A \mid C$ No unmeasured A-M confounding

iv) $Y(a, m) \perp\!\!\!\perp M(a^*) \mid C$ cross-world exchangeability

(fail)

M-Y confounder affected by exposure

NPSEM



$$\star Y(a, m) : A = a, M = m$$

$$L(a) = g_L(a, E_L)$$

$$Y(a, m) = g_Y(a, L(a), m, E_Y)$$

$$= g_Y(a, g_L(a, E_L), m, E_Y)$$

$$= f_Y(E_L, E_Y)$$

$$A = g_A(E_A)$$

$$L = g_L(A, E_L)$$

$$M = g_M(A, L, E_M)$$

$$Y = g_Y(A, L, M, E_Y)$$

$$\star M(a^*) : A = a^*, L(a^*) = g_L(a^*, E_L)$$

$$M(a^*) = g_M(a, L(a^*), E_M) = g_M(a, g_L(a^*, E_L), E_M)$$

$$= f_M(E_L, E_M)$$

★ Mediation formula

$$E[Y(a, M(a^*))] = \int_C E[Y(a, M(a^*)) | C=c] P(C=c)$$

$$= \int_C \left\{ \int_m E[Y(a, M(a^*)) | M(a^*)=m, C=c] P(M(a^*)=m | C=c) \right\} P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y(a, m) | M(a^*)=m, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp M(a^*) | C) = \int_C \int_m E[Y(a, m) | C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp A | C) = \int_C \int_m E[Y(a, m) | A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp M | A, C) = \int_C \int_m E[Y(a, m) | M=m, A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(M(a^*) \perp\!\!\!\perp A | C) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M(a^*)=m | A=a^*, C=c) P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M=m | A=a^*, C=c) P(C=c)$$

$$= Q(a, a^*)$$

$$\phi(a, a^*) \triangleq Q(a, a^*)$$

identify

$$TE = NDE + NIE$$

$$\hat{\phi}(1,1) - \hat{\phi}(0,0) = [\hat{\phi}(1,0) - \hat{\phi}(0,0)] + [\hat{\phi}(1,1) - \hat{\phi}(1,0)]$$

✍ Estimation

Two traditional approaches to mediation analysis

i) The Difference Method

ii) The Product Method

- ★ The difference method has been employed more frequently in epidemiology and biomedical sciences. It consists of two regression models.

$$Y = \beta_0 + \beta_1 A + \epsilon_1$$

$$Y = \theta_0 + \theta_1 A + \theta_2 M + \epsilon_2$$

The exposure coef θ_1 , in the model that includes the mediator is then generally take as a measure of DE

$$\begin{aligned} DE &= \theta_1 \\ TE &= \beta_1 \end{aligned} \Rightarrow IE = \beta_1 - \theta_1$$

- ★ A slightly different method, sometimes called the product-of-coefficients method, is used w/ more frequency in the social sciences.

Two regressions are employed.

$$\begin{aligned} Y &= \theta_0 + \theta_1 A + \theta_2 M + \epsilon_2 \\ M &= \alpha_0 + \alpha_1 A + \epsilon_3 \end{aligned}$$

the product $\alpha_1 \theta_2$, taken as a measure of indirect effect

$$IE = \alpha_1 \theta_2$$

- ★ Fortunately, for $\begin{cases} Y: \text{conti} \\ M: \text{conti} \end{cases}$ w/ linear regression fit by OLS

the two approaches are coincide.

$$\beta_1 - \theta_1 = \alpha_1 \theta_2$$

Recall the identification of $\phi(a, a^*) = E[Y(a, M(a^*))]$

$$\int_c \int_m E(Y | M=m, A=a, C=c) P(M=m | A=a^*, C=c) P(C=c)$$

$$\text{or } \int_c \int_m E(Y | M=m, A=a, C=c) f_{M|A,C}(m | a^*, c) f_C(c) dm dc$$

(i) Model Y:

$$E(Y|A, M, C) = \theta_0 + \theta_1 A + \theta_2 M + \theta_c C$$

(ii) Model M:

$$E(M|A, C) = \alpha_0 + \alpha_1 A + \alpha_c C$$

$$\begin{aligned} \Rightarrow Q(a, \bar{a}) &= \int_c \int_m (\theta_0 + \theta_1 a + \theta_2 m + \theta_c C) f_{M|A, C}(m|a^*, c) f_c(c) dm dc \\ &= \theta_0 + \theta_1 a + \theta_2 \int_c \int_m m f_{M|A, C}(m|a^*, c) dm f_c(c) dc + \theta_c E(C) \end{aligned}$$

$$B = \theta_2 \int_c E(M|A=a^*, C=c) f_c(c) dc$$

$$= \theta_2 \int_c (\alpha_0 + \alpha_1 a^* + \alpha_c C) f_c(c) dc$$

$$= \theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c E(C))$$

then

$$Q(a, \bar{a}) = \theta_0 + \theta_1 a + \theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c E(C)) + \theta_c E(C)$$

$$\theta_1 a + \theta_2 \alpha_1 a^* + \text{Const}$$

$$NDE = \phi(1, 0) - \phi(0, 0) = Q(1, 0) - Q(0, 0) = \theta_1 \rightarrow \text{DE in difference method}$$

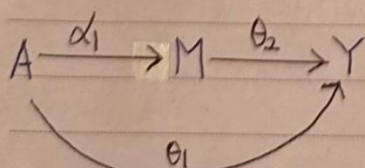
$$NIE = \phi(1, 1) - \phi(1, 0) = Q(1, 1) - Q(1, 0) = \theta_2 \alpha_1$$

$\rightarrow$ IE in product method

$$(iii) E(Y|A, C) = \theta_0 + \theta_1 A + \theta_c C$$

$$\Downarrow$$

$$\begin{aligned} E_M(E(Y|M, A, C)) &= E_M(\theta_0 + \theta_1 A + \theta_2 M + \theta_c C) \\ &= \theta_0 + \theta_1 A + \theta_2 E_M(M|A, C) + \theta_c C \\ &= \theta_0 + \theta_1 A + \theta_2 (\alpha_0 + \alpha_1 A + \alpha_c C) + \theta_c C \\ &= (\theta_0 + \theta_2 \alpha_0) + (\theta_1 + \theta_2 \alpha_1) A + (\theta_c + \theta_2 \alpha_c) C \end{aligned}$$



Difference Method

$$\theta_1 = \theta_1 + \theta_2 \alpha_1$$

Product Method

★ Exposure - Mediator Interaction

$$\text{Model } Y : E(Y|M, A, C) = \theta_0 + \theta_1 A + \theta_2 M + \theta_3 \underbrace{AM}_{\substack{\text{A-M} \\ \text{interaction}}} + \theta_4 C$$

Recall

$$Q(a, a^*) = \int_C \int_m E(Y|M=m, A=a, C=c) f_{M|A,C}(m|a^*, c) f_C(c) dm dc$$

$$= \theta_0 + \theta_1 a + \underbrace{\int_C \int_m (\theta_2 + \theta_3 a) m f_{M|A,C}(m|a^*, c) f_C(c) dm dc}_{\substack{\parallel \\ (\Delta)}} + \theta_4 E(C)$$

$$(\Delta) = (\theta_2 + \theta_3 a) \int_C E(M|A=a^*, C=c) f_C(c) dc$$

$$= (\theta_2 + \theta_3 a) E(\alpha_0 + \alpha_1 a^* + \alpha_C C)$$

$$= (\theta_2 + \theta_3 a) [\alpha_0 + \alpha_1 a^* + \alpha_C E(C)]$$

then

$$Q(a, a^*) = (\theta_0 + \theta_1 a) + (\theta_2 + \theta_3 a) [\alpha_0 + \alpha_1 a^* + \alpha_C E(C)] + \theta_4 E(C)$$

$$= \theta_0 + \theta_2 \alpha_0 + \theta_2 \alpha_C E(C) + \theta_4 E(C)$$

$$+ (\theta_1 + \alpha_0 \theta_3 + \alpha_C \theta_3 E(C)) a$$

$$+ \theta_2 \alpha_1 a^*$$

$$+ \theta_3 \alpha_1 a a^*$$

$$\triangleq \text{Const} + Q_1 a + Q_2 a^* + Q_3 a a^*$$

$$NIE = Q(1, 1) - Q(1, 0) = (Q_1 + Q_2 + Q_3) - Q_1 = \alpha_1 (\theta_2 + \theta_3)$$

$$NDE = Q(1, 0) - Q(0, 0) = Q_1 = \theta_1 + \theta_3 (\alpha_0 + \alpha_C E(C))$$

★ MC Estimation

$$Q(a, a^*) = \int_C \int_m E(Y|M=m, A=a, C=c) P(M=m|A=a^*, C=c) P(C)$$

$$= \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$$

$$E(Y|M=m, A=a, C=c) = \theta_0 + \theta_1 a + \theta_2 m + \theta_4 C = g_Y(a, m, C; \theta)$$

(+ $\theta_3 am + \theta_4 C + \theta_5 mc$ if consider interaction)

$$\hat{Y}_i = g_Y(A_i, \hat{M}_i, \hat{C}_i; \hat{\theta})$$

$$\hat{C}_i \sim P(C=c)$$

$$\hat{M}_i \sim \hat{P}(M=m|A=a^*, \hat{C}_i)$$

Id C_i A_i M_i Y_i 1. Build model for Y, M

$$E(Y|M, A, C) = g_Y(A, M, C, \theta)$$

$$E(M|A, C) = g_M(A, C, \alpha)$$

2. $\hat{\alpha}, \hat{\Delta M}, \hat{\theta}$

by REML

(D1) id C A M $Y \leftarrow Y(1) = Y(1, M(1))$

1
2
...
n

$$C_i \sim \hat{P}(C=c), A=1$$

$$M \sim g_M(A=1, C, \hat{\alpha}, \hat{\Delta M})$$

$$Y \sim g_Y(A=1, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(1)) = \hat{\phi}(1, 1)$$

(D2) id C A M $Y \leftarrow Y(0) = Y(0, M(0))$

1
2
...
n

$$C_i \sim \hat{P}(C=c), A=0$$

$$M \sim g_M(A=0, C, \hat{\alpha}, \hat{\Delta M})$$

$$Y \sim g_Y(A=0, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(0)) = \hat{\phi}(0, 0)$$

(D3) id C A M $Y \leftarrow Y(1, M(0))$

1
2
...
n

$$C_i \sim P(C=c), A=1$$

$$Y \sim g_Y(A=1, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(1, M(0))) = \hat{\phi}(1, 0)$$

統計

流行病學

JASA, JRRSB, Biometrika, Annals $\Leftrightarrow$ Epidemiology, AJE, IJE, EJE